Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the accuracy during training.

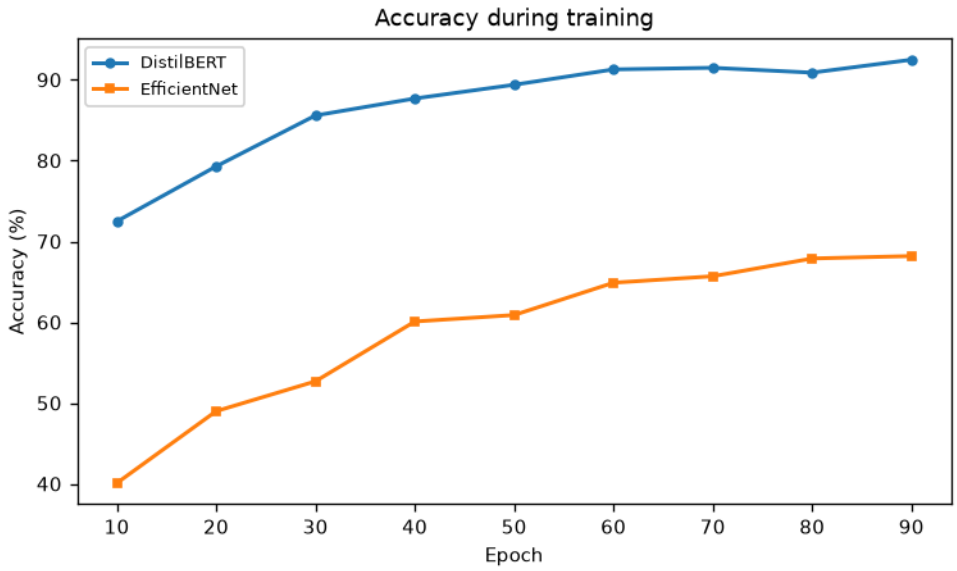


Figure 1: Accuracy during training for the models in this experiment.

EfficientNet reaches 60.1% accuracy at epoch 40. EfficientNet reaches 94.5% accuracy at epoch 70. As shown in Figure 1, EfficientNet reaches 65.7% accuracy at epoch 70.

Figure 2 shows the f1 score during training.

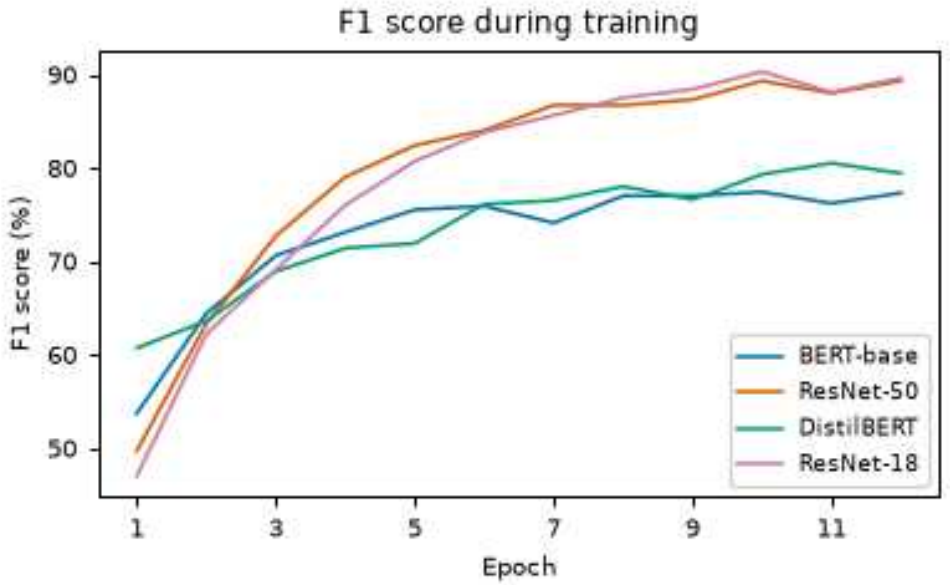


Figure 2: F1 score during training for the models in this experiment.

At epoch 11, BERT-base achieves 76.2% F1 score. As shown in Figure 2, BERT-base reaches 70.6% F1 score at epoch 3. As shown in Figure 2, DistilBERT attains a F1 score of 54.1% at epoch 2. ResNet-50 reaches 77.4% F1 score at epoch 10.

Figure 3 shows the f1 score by dataset.

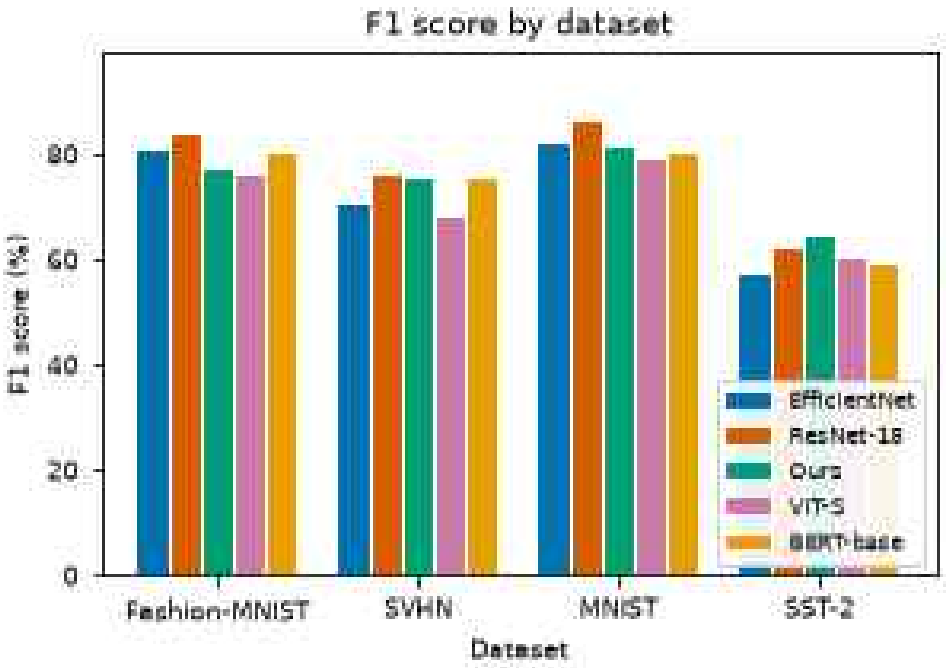


Figure 3: F1 score by dataset for the models in this experiment.

On SVHN, ViT-S obtains a F1 score of 68.1%. BERT-base achieves 82.1% F1 score on MNIST. ResNet-18 achieves over 70% F1 score on SVHN (Figure 3).

Figure 4 shows the validation accuracy by dataset.

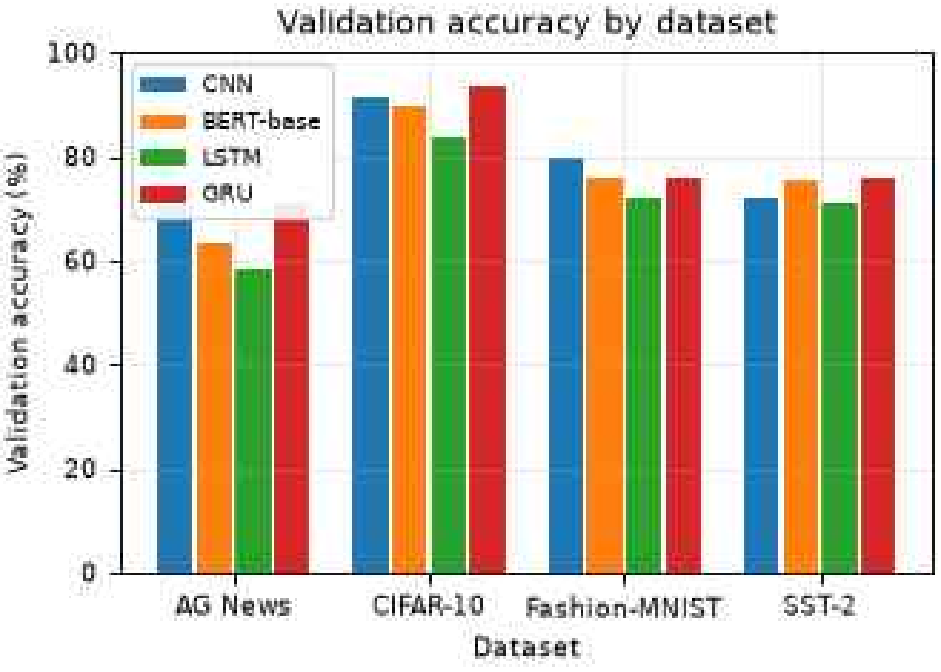


Figure 4: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 4, CNN outperforms BERT-base by 21.5 percentage points on CIFAR-10. On AG News, LSTM obtains a validation accuracy of 53.1%. As shown in Figure 4, CNN achieves 66.0% validation accuracy on AG News. GRU achieves 94% validation accuracy on CIFAR-10 (Figure 4).

3 Discussion

On Tiny-ImageNet, XGBoost obtains an accuracy of 81.2%. We note that the training was stable for all

models.